$$x \diamond (x \diamond x) \;=\; y \diamond (y \diamond y)$$
$$(x \diamond x) \diamond x \;=\; (y \diamond y) \diamond y$$

The equational theory with maximum variance and Y-value is $\mathsf{Eqn}[3658]$ which is self-conjugate:

$$x \diamond x \;=\; (x \diamond x) \diamond (x \diamond x)$$

6.6 The Z-axis detects conjugacy

One immediate and striking observation on the latent space is the Z-value produces a reflection symmetry between conjugate theories, around the plane $Z = 0$. We noticed this emerging reflection symmetry between conjugate theories in our first experiments, and improved this reflection symmetry by selecting a perfectly symmetric sample of finite magmas. For reference, the equational theory with maximum Z-value is $\mathsf{Eqn}[1022]$

$$x \;=\; x \diamond ((x \diamond (x \diamond y)) \diamond y)$$

closely followed by $\mathsf{Eqn}[428]$

$$x \;=\; x \diamond (y \diamond (x \diamond (y \diamond x)))$$

By reflection symmetry, the theory with minimum Z-value is the conjugate $\mathsf{Eqn}[2529]$:

$$x \;=\; (y \diamond ((y \diamond x) \diamond x) \diamond x$$

closely followed by the conjugate $\mathsf{Eqn}[3067]$:

$$x \;=\; (((x \diamond y) \diamond x) \diamond y) \diamond x$$

Accordingly, there are 84 self-conjugate equational theories and they all appear on the plane $Z = 0$. This ability of the Z-component to detect conjugacy is remarkable since two conjugate theories $\mathsf{Eqn}[j]$ and $\mathsf{Eqn}[k]$ have the same Stone spectrum, and thus the same expectation and variance, see the definition in §5.3 and discussion in §6.3.

7 Clustering of provably equivalent theories in the latent space

Now that we have constructed the latent space of equational theories, we can embed the implication graph $(G, \rightsquigarrow)$ to visualize what it looks like in three dimensions. For every pair of equational theories related by a proof $\mathsf{Eqn}[j] \Longrightarrow \mathsf{Eqn}[k]$ we thus draw an *edge* $\mathsf{Eqn}[j] \longrightarrow \mathsf{Eqn}[k]$ between the vertices associated to $\mathsf{Eqn}[j]$ and $\mathsf{Eqn}[k]$ in the latent space. We use the same terminology as for implications, and thus distinguish three classes of edges: the *reversible* edges $\sim$, the *atomic* edges $\rightsquigarrow$ which are strict by definition, and the *strict* edges $\Rightarrow$ which are general and not necessarily atomic. The length of an edge can be computed in several ways. In this paper, we choose the simplest distance which is the Euclidian distance in the latent space.

At this stage, it is important to notice that two provably equivalent theories $\mathsf{Eqn}[j]$ and $\mathsf{Eqn}[k]$ may induce different Stone pairings within a given finite magma $(A, \diamond)$. Indeed, $\mathsf{Eqn}[j] \sim \mathsf{Eqn}[k]$ simply ensures that validity is preserved

$$(A, \diamond) \vDash \mathsf{Eqn}[j] \iff (A, \diamond) \vDash \mathsf{Eqn}[k]$$

in other words, that the Stone pairing of $\mathsf{Eqn}[j]$ is equal to 1 if and only if the Stone pairing of $\mathsf{Eqn}[k]$ is equal to 1. On the other hand, we observe empirically that the reversible edges are on average seven times shorter than strict atomic edges, and ten times shorter than general stric edges, as indicated in the table below.

	Reversible	**Atomic**	**Strict**
count	2 470 916	1 052 209	5 707 363
mean	0.69	5.29	7.17
std	0.65	2.22	4.30
min	0.00	0.00	0.00
25%	0.00	4.14	4.17
50%	0.63	4.69	6.14
75%	0.96	6.33	9.22
max	5.36	37.08	41.20

This important difference suggests a degree of clustering of the provably equivalent theories in the latent space. The hypothesis can be validated by computing the center of mass of each clique of provably equivalent theories, and connecting each theory to the center of mass of its corresponding clique. One obtains in this way a three-dimensional picture, see Fig. 6, where the clusters of provably equivalent theories are clearly separated.

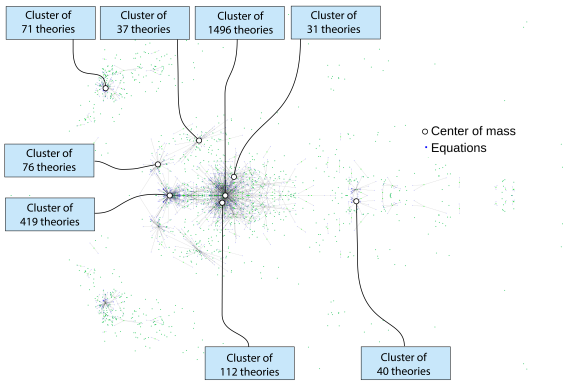

Figure 6: Visualizing the center of mass of each provably equivalent clique. Each equational theory is drawn connected to its corresponding center of mass.

8 Implication flows in the latent space

The clustering of provably equivalent theories in the latent space observed in §7 has an important